Single-Pass Estimation of the Clustering Coefficient Distribution in Graph Streams: Supplementary Material

Cristian Boldrin*
University of Padova
Padova, Italy
boldrincri@dei.unipd.it

C. Seshadhri
University of California, Santa Cruz
Santa Cruz, California, USA
sesh@ucsc.edu

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1 OUTLINE

The Supplementary Material is organized as follows. TODO: outline.

2 TOOLS

In the following, we report concentration inequalities used for deriving our results.

Lemma 2.1 (Multiplicative Chernoff Bound). *Let $X_1, \ldots, X_n$ be independent random variables taking values in $[0,1]$. Let $X = \sum_{i=1}^n X_i$, and let $\mu = \mathbb{E}[X]$. Then, for any $\delta \in [0,1]$,*

$$\mathbb{P}\left[|X-\mu| \geq \delta\mu\right] \leq 2\exp\left(\frac{-\delta^2\mu}{3}\right).$$

Lemma 2.2 (Hoeffding's Inequality). *Let $X_1, \ldots, X_n$ be random variables with $X_i \in [0,1]$ that are either independent or negatively associated. Let $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$, and $\mu = \mathbb{E}\left[\bar{X}\right]$. Then, for any $t \geq 0$,*

$$\mathbb{P}\left[\left|\bar{X}-\mu\right| \geq t\right] \leq 2\exp\left(-2nt^2\right)$$

Lemma 2.3 (Bernstein's Inequality). *Let $X_1, \ldots, X_n$ be independent random variables such that $|X_i - \mathbb{E}[X_i]| \leq M$ for all i. Let $S = \sum_{i=1}^n X_i$, and $V = \sum_{i=1}^n \mathrm{Var}[X_i]$. Then, for any $t \geq 0$,*

$$\mathbb{P}\left[|S-\mathbb{E}[S]| \geq t\right] \leq 2\exp\left(-\frac{t^2}{2V+\frac{2}{3}Mt}\right)$$

Lemma 2.4 (One Sided Chernoff Like Bound). *Let $X_1, ..., X_n$ be independent random variables such that each $X_i \sim Bern(p)$ with $p < \tau(1-\varepsilon)$ for some $\varepsilon \in (0, 1/2]$. Let $X = \sum_{i=1}^n X_i$. Then, we have:*

$$\mathbb{P}\left[X \geq n\tau\right] \leq \exp\left(\frac{-\varepsilon^2 n\tau}{10}\right).$$

Proof. The proof follows immediately from Lemma 22 of [1], where instead of showing concentration on the average $\frac{1}{n}\sum_{i=1}^n X_i$ (as done in [1]), we consider the sum $X = \sum_{i=1}^n X_i$. □

In the following, we leverage a result from [4] in the context of dependency graphs. For random variables $X_1, \ldots, X_n$, a graph H with vertex set $V = \{1, \ldots n\}$ is called a *dependency graph* [3] for variables X_i's if, for any two disjoint subsets $A, B \subset V$ such that there are no edges connecting a vertex in A to a vertex in B, the collections $\{X_i\}_{i\in A}$ and $\{X_j\}_{j\in B}$ are mutually independent.

Lemma 2.5 (Bernstein-type ineq. for dependency graphs [4, Thm. 2.3]). *Let $X_1, \ldots, X_n$ be random variables with a dependency graph H and $\mathbb{E}[X_i] = 0$ for all i. Assume $|X_i| \leq b$ for all i, and let $V = \sum_{i=1}^n \mathrm{Var}[X_i]$. Let $\chi^*(H)$ be the fractional chromatic number of H. Then for all $t \geq 0$,*

$$\mathbb{P}\left[\left|\sum_{i=1}^n X_i\right| \geq t\right] \leq 2\exp\left(-\frac{8t^2}{25\chi^*(H)\left(V+\frac{bt}{3}\right)}\right).$$

Above, the fractional chromatic number $\chi^*(H)$ is defined as the linear programming relaxation of the standard chromatic number $\chi(H)$. Intuitively, it represents the minimum weight needed to cover the vertices of H with independent sets. It holds $\chi^*(H) \leq \chi(H) \leq d_{max}(H) + 1$, where $d_{max}(H)$ is the maximum degree for graph H.

3 ADDITIONAL PROOFS

In the following, we state all our derived results together with their proofs.

Lemma 3.1. *Let $\widehat{d}_u$ be the estimated degree for each node u contained in node-degree maps S_h and S_t returned by* `FirstPass` *algorithm, using fixed probability p_t for sampling edges. Then:*

(i) $\widehat{d}_u = d_u$ for each node $u \in S_h$,

and, for any $\varepsilon \in (0, 1/2)$, if $\tau = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2 p_t}\right)$, we have, with high probability:

(ii) if $d_v \leq (1-\varepsilon)\tau$, then $\widehat{d}_v \leq \tau$ for each node $v \in S_t$;
(iii) if $d_v \geq \tau$, then $\widehat{d}_v \in (1 \pm \varepsilon)d_v$ for each node $v \in S_t$;
(iv) if $d_v \geq \tau$, then $v \in E_t$.

Proof. *Item (i).* By design, `FirstPass` is tracking exactly degrees of nodes in S_h, which are sampled via hashing.

Item (ii). Fix a vertex v. Let $X_i(v)$ be 1 if the i-th edge of the stream is sampled and incident to v, and 0 otherwise, for $i = 1, ..., m$. At the end of the stream, we have that $X_i(v) \sim Bern\left(p_t\, d_v/m\right)$, with $X_i(v)$'s independent. We compute $\widehat{d}_v = 1/p_t \sum_{i=1}^m X_i(v)$. Since

*Work done in part while visiting University of California, Santa Cruz.

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$d_v \leq (1-\varepsilon)\tau$, we can apply Lemma 2.4:

$$\mathbb{P}\left[\widehat{d}_v > \tau\right] = \mathbb{P}\left[\sum_{i=1}^{m} X_i(v) \geq m\ \frac{p_t \tau}{m}\right] \leq \exp\left(\frac{-\varepsilon^2 p_t \tau}{10}\right),$$

which is $\leq 1/n^c$ for the suggested choice of τ. Item follows then by a union bound over all degrees $d \leq (1-\varepsilon)\tau$ and vertices $v \in S_t$.

Item (iii). Fix a degree $d \geq \tau$ and a vertex v. Again, let $X_i(v)$ be 1 if the i-th edge of the stream is sampled and is incident to v, and 0 otherwise, for $i = 1, ..., m$. We have $\mathbb{E}\left[X_i(v)\right] = pt\ d_v/m$. From line **??**, we compute $\widehat{d}_v = 1/p_t \sum_{i=1}^{m} X_i(v)$. Thus, $\mathbb{E}\left[\widehat{d}_v\right] = d_v \geq \tau$. By Chernoff bound 2.1, we have:

$$\mathbb{P}\left[\left|\widehat{d}_v - d_v\right| \geq \varepsilon d_v\right] \leq 2\exp\left(\frac{-\varepsilon^2 p_t \tau}{3}\right),$$

which is $\leq 1/n^c$ for the suggested choice of τ. Item follows then by a union bound over all degrees $d \geq \tau$ and vertices v with $d_v \geq \tau$.

Item (iv). Fix a vertex v such that $d_v \geq \tau$. Again, let $X_i(v)$ be 1 if the i-th edge of the stream is sampled in E_t, and is incident to v. We have $X_v = \sum_{i=1}^{m} X_i(v) \sim Bin(m, p_v)$, where $p_v = p_t \frac{d_v}{m} \geq p_t \frac{\tau}{m}$. The probability that vertex v will not appear in E_t is the probability of X_v being 0, that is $(1-p_v)^m \leq \exp(-p_t\tau) \leq 1/n^{\varepsilon^{-2}} \leq 1/n^c$ for the above choice of τ. Item follows then by a union bound over vertices v with $d_v \geq \tau$. □

Fact 3.2. *Given $\widehat{d}$ such that $\widehat{d} \in (1 \pm \varepsilon)\, d$, and $d \geq \tau$, it holds:*

$$\binom{\widehat{d}}{2} \in \left[(1-\varepsilon)^2 - \frac{\varepsilon(1-\varepsilon)}{\tau-1}, (1+\varepsilon)^2 + \frac{\varepsilon(1+\varepsilon)}{\tau-1}\right]\binom{d}{2}.$$

Corollary 3.3. *Following Fact 3.2, for any $\varepsilon \in (0, 1/2], \tau \geq 4$:*

$$\binom{\widehat{d}}{2} \in (1 \pm 3\varepsilon)\binom{d}{2}.$$

Proof. From Fact 3.2, we analyze the upper and lower bound, separately. For the upper bound, since $\varepsilon \leq 1/2$, we have $\varepsilon^2 \leq \varepsilon/2$. Thus:

$$(1+\varepsilon)^2 + \frac{\varepsilon(1+\varepsilon)}{\tau-1} = 1 + 2\varepsilon + \varepsilon^2 + \frac{\varepsilon(1+\varepsilon)}{\tau-1} \leq 1 + 2.5\varepsilon + \frac{1.5\varepsilon}{\tau-1}.$$

For $\tau \geq 4$, it follows an upper bound of $1 + 3\varepsilon$. For the lower bound:

$$(1-\varepsilon)^2 - \frac{\varepsilon(1-\varepsilon)}{\tau-1} = 1 - 2\varepsilon + \varepsilon^2 - \frac{\varepsilon(1-\varepsilon)}{\tau-1} \geq 1 - 2\varepsilon - \frac{\varepsilon}{\tau-1}.$$

For $\tau \geq 4$, the expression is $> 1 - 3\varepsilon$. □

Notice that in our experiments, we have values of τ from 253 (as-skitter) and up to 4448 (twitter).

Quality of Estimates. In the following, we report proofs of results analyzing the quality of the estimates for binned-degree clustering coefficients, returned by our proposed (two-pass) approach. Note that such results extend to our one-pass approach (see Section **??**).

We start by analyzing the head estimator. We prove Theorem **??** of main paper by showing bounds for NDCC and WDCC separately. First, we provide concentration for estimates of the number of nodes and number of triangles. Throughout, we assume degree interval $D = [L, U)$, with $U = \Theta(L)$.

Lemma 3.4. *Consider degree interval D and any $\varepsilon \in (0, 1)$. Let $C_h(D)$ be the number of nodes with degree in D in the sample S_h. Then, if $p_h = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2\, \mathrm{C}(D)}\right)$, we have, with high probability, $C_h(D) \in (1 \pm \varepsilon) p_h\, \mathrm{C}(D)$.*

Proof. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. Note that X_i's are independent. We have $C_h(D) = \sum_{u, d_u \in D} X_u$, and $\mathbb{E}\left[C_h(D)\right] = p_h\, \mathrm{C}(D)$. Result follows from an application of Lemma 2.1. □

We show concentration bounds also for the number of triangles incident to sampled nodes. Recall that $\Delta_V(D)$ is the maximum number of triangles incident to a node with degree in D.

Lemma 3.5. *Consider degree interval D and any $\varepsilon \in (0, 1)$. Let $T_h(D)$ be the number of triangles incident to nodes with degree in D in the sample S_h. Then, if $p_h = \widetilde{\Omega}\left(\frac{\Delta_V(D)}{\varepsilon^2\, \mathrm{T}(D)}\right)$, we have, with high probability, $T_h(D) \in (1 \pm \varepsilon) p_h\, \mathrm{T}(D)$.*

Proof. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. Note that X_i's are independent. We have $T_h(D) = \sum_{u, d_u \in D} X_u t_u$, and $\mathbb{E}\left[T_h(D)\right] = p_h\, \mathrm{T}(D)$. By definition, it holds $X_u t_u \leq \Delta_V(D)$, for each u. Result follows by an application of Lemma 2.1 to the normalized variables $X_u t_u / \Delta_V(D)$. □

Theorem 3.6. *Consider degree interval $D = [L, U)$ such that $L < U \leq \tau$. For any $\varepsilon, \eta \in (0, 1)$, if*

$$p_h = \widetilde{\Omega}\left(\frac{1}{\eta^2}\left(\frac{1}{\mathrm{C}(D)} + \frac{\Delta_V(D)}{\mathrm{T}(D)}\right)\right) \text{ and } p_{M_h} p_{A_h} = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2 p_h\, \mathrm{T}(D)}\right),$$

*our head estimator from Alg. **??** outputs a value $\widehat{NDCC}(D)$ such that:*

$$\left|\widehat{NDCC}(D) - \mathrm{NDCC}(D)\right| \leq \varepsilon \cdot \mathrm{NDCC}(D) + \eta.$$

with high probability.

Proof. Recall that $\widehat{d}_u = d_u$ for any $u \in S_h$, since degrees are tracked exactly (Lemma 3.1, item (i)). Our head estimator from Alg. **??** computes estimates as:

$$\widehat{NDCC}(D) = \frac{1}{|\{u \in S_h, d_u \in D\}|} \sum_{u \in S_h, d_u \in D} \frac{\widehat{t}_u}{\binom{d_u}{2}}. \tag{1}$$

Let $\widehat{C}(D) = \left|\left\{u \in S_h, d_u \in D\right\}\right|$. Define the quantity:

$$Y(D) = \frac{1}{\widehat{C}(D)} \sum_{u \in S_h, d_u \in D} \frac{t_u}{\binom{d_u}{2}}$$

We bound the overall error as follows:

$$\begin{aligned} &|\widehat{NDCC}(D) - \mathrm{NDCC}(D)| \\ &\leq \underbrace{|\mathrm{NDCC}(D) - Y(D)|}_{(i)\text{: node sampling error}} + \underbrace{|\widehat{NDCC}(D) - Y(D)|}_{(ii)\text{: triangle estimator error}}. \end{aligned} \tag{2}$$

First, we bound term (i) of Eq. 2. Let $Z_u = t_u / \binom{d_u}{2}$ if $u \in S_h$, and 0 otherwise; we have $Z_u \in [0, 1]$ and $Y(D) = \sum_{u, d_u \in D} Z_u / \widehat{C}(D)$. Conditioning on the event $\widehat{C}(D) = C \in (1 \pm \eta) p_h\, \mathrm{C}(D)$ (see Lemma 3.4), we have that variables Z_i's are negatively associated. That is, $\mathbb{P}\left[Z_u \neq 0 \mid \widehat{C}(D) = C\right] = C/\mathrm{C}(D)$. Therefore:

$$\mathbb{E}\left[Y(D) \,\middle|\, \widehat{C}(D) = C\right]$$

$$= \frac{1}{C} \sum_{u,d_u \in D} \frac{t_u}{\binom{d_u}{2}} \mathbb{E}\left[u \in S_h, d_u \in D \,\middle|\, \widehat{C}(D) = C\right]$$
$$= \frac{1}{C} \sum_{u,d_u \in D} \frac{t_u}{\binom{d_u}{2}} \frac{C}{\mathrm{C}(D)} = \mathrm{NDCC}(D)\,.$$

Applying Hoeffding's inequality (Lemma 2.2) for negative associated variables:

$$\mathbb{P}\left[|Y(D) - \mathrm{NDCC}(D)| \geq \eta \,\middle|\, \widehat{C}_h = C\right]$$
$$= \mathbb{P}\left[\left|\frac{\sum_{u,d_u \in D} Z_u}{C} - \mathrm{NDCC}(D)\right| \geq \eta \,\middle|\, \widehat{C}_h = C\right]$$
$$\leq 2\exp\left(-2C\eta^2\right) \leq 2\exp\left(-2(1-\eta)p_h\,\mathrm{C}(D)\eta^2\right),$$

which is $\leq 1/n^c$ for $p_h = \widetilde{\Omega}\left(\frac{1}{\eta^2\,\mathrm{C}(D)}\right)$, which bounds the first term of Eq. 2. Applying union bound over both events concludes the proof for case 1.

Now, we bound term (ii) of Eq. 2, which controls the error of our triangle-estimator. Our goal boils down to bounding:

$$\frac{1}{\widehat{C}(D)} \left| \sum_{u \in S_h, d_u \in D} \frac{\widehat{t}_u - t_u}{\binom{d_u}{2}} \right|. \tag{3}$$

Define $\widehat{A}(D), A(D)$ as $\sum_{u \in S_h, d_u \in D} \frac{\widehat{t}_u}{\binom{d_u}{2}}$ and $\sum_{u \in S_h, d_u \in D} \frac{t_u}{\binom{d_u}{2}}$, respectively. Eq. 3 can be expressed as $\frac{1}{\widehat{C}(D)}\left|\widehat{A}(D) - A(D)\right|$. Let $\mathcal{T}_h(D)$ be the set of triangles incident to nodes in S_h, having degree in D. We have $|\mathcal{T}_h(D)| = \mathrm{T}_h(D)$. For each triangle Δ, define:

$$w_\Delta = \sum_{u \in \Delta \cap S_h, d_u \in D} \frac{1}{\binom{d_u}{2}} \in \left[\frac{1}{\binom{U}{2}}, \frac{3}{\binom{L}{2}}\right].$$

We have $\sum_{\Delta \in \mathcal{T}_h(D)} w_\Delta = A(D)$. Let $I_\Delta = 1$ if triangle Δ is counted by Alg. **??**, and 0 otherwise. We can write:

$$\widehat{A}(D) = \sum_{\Delta \in \mathcal{T}_h(D)} \frac{w_\Delta}{p_\Delta} I_\Delta, \text{ and } \mathbb{E}\left[\widehat{A}(D)\right] = A(D).$$

Let $X_\Delta = \frac{w_\Delta I_\Delta}{p_\Delta}$. We have $\mathrm{Var}\left[X_\Delta\right] \leq \frac{9p_\Delta(1-p_\Delta)}{\binom{L}{2}^2 p_\Delta^2} \leq \frac{9}{\binom{L}{2}^2 p_A p_M}$, assuming $p_A \leq p_M$. Define centerd variables $Y_\Delta = X_\Delta - \mathbb{E}\left[X_\Delta\right]$. We have $Y_\Delta = w_\Delta\left(\frac{I_\Delta}{p_\Delta} - 1\right) \leq \frac{3}{\binom{L}{2} p_M p_A}$. Note that $\sum_\Delta X_\Delta = \widehat{A}(D)$.

Consider the dependency graph H for variables $X_1, \ldots X_{T_h(D)}$. That is, place a vertex in H for every variable X_i, and place an edge between X_{Δ_i} and X_{Δ_j} if they are dependent, i.e., if triangle Δ_i and triangle Δ_j share a sampled edge during the streaming algorithm. The fractional chromatic number $\chi^*(H)$ is $\leq \Delta_E(D) + 1$, where $\Delta_E(D)$ is the maximum number of triangles incident to a single edge which contains a node with degree in D, in the original graph G. Note that, by definition of graph H, we have $\Delta_E(D) = d_{max}(H)$

Moreover, $\sum_\Delta \mathrm{Var}\left[Y_\Delta\right] = \sum_\Delta \mathrm{Var}\left[X_\Delta\right] \leq \mathrm{T}_h(D)\,\frac{9}{\binom{L}{2}^2 p_A p_M}$. By defining $Y = \sum_\Delta Y_\Delta$, we can apply Lemma 2.5:

$$\mathbb{P}\left[\left|\widehat{A}(D) - A(D)\right| \geq \varepsilon A(D)\right] = \mathbb{P}\left[|Y| \geq \varepsilon A(D)\right]$$
$$\leq 2\exp\left(\frac{-8\varepsilon^2 A^2(D)}{25\Delta_E(D)\left(\frac{9\,\mathrm{T}_h(D)}{\binom{L}{2}^2 p_M p_A} + \frac{\varepsilon A(D)}{\binom{L}{2} p_M p_A}\right)}\right),$$

which is $\leq 1/n^c$ given that $A(D) \in T_h(D)\left[1/\binom{U}{2}, 3/\binom{L}{2}\right]$, $U/L = \Theta(1)$, for $p_M p_A = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2 T_h(D)}\right)$. We then use Lemma 3.5 to obtain concetration for $T_h(D)$ around $p_h\,\mathrm{T}(D)$, Thus, we showed:

$$\left|\widehat{NDCC}(D) - Y(D)\right| = \frac{1}{\widehat{C}(D)}\left|\widehat{A}(D) - A(D)\right| \leq \frac{\varepsilon A(D)}{\widehat{C}(D)} = \varepsilon Y(D),$$

with high probability. Combining item (i), (ii) we proved, for $D = [L, U)$ with $L \leq U \leq \tau$, that:

$$\left|\widehat{NDCC}(D) - \mathrm{NDCC}(D)\right| \leq \eta + \varepsilon Y(D) \leq \eta + \varepsilon(\mathrm{NDCC}(D) + \eta)$$
$$= \varepsilon \cdot \mathrm{NDCC}(D) + (1+\varepsilon)\eta.$$

Result from the theorem follows then by rescaling η.

□

Theorem 3.7. *Consider degree interval $D = [L, U)$ such that $L < U \leq \tau$. For any $\varepsilon, \eta \in (0,1)$, if*

$$p_h = \widetilde{\Omega}\left(\frac{1}{\eta^2}\left(\frac{1}{\mathrm{C}(D)} + \frac{\Delta_V(D)}{\mathrm{T}(D)}\right)\right) \text{ and } p_{M_h} p_{A_h} = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2 p_h\,\mathrm{T}(D)}\right),$$

our head estimator from Alg. **??** *outputs a value $\widehat{WDCC}(D)$ such that:*

$$\left|\widehat{WDCC}(D) - \mathrm{WDCC}(D)\right| \leq \varepsilon \cdot \mathrm{WDCC}(D) + \eta.$$

Proof. Recall that $\widehat{d}_u = d_u$ for any $u \in S_h$, since degrees are tracked exactly (Lemma 3.1, item (i)). Our head estimator from Alg. **??** computes estimates as:

$$\widehat{WDCC}(D) = \frac{\sum_{u \in S_h, d_u \in D} \widehat{t}_u}{\sum_{u \in S_h, d_u \in D} \binom{d_u}{2}}. \tag{4}$$

Define the quantity using true triangle counts:

$$Y(D) = \frac{\sum_{u \in S_h, d_u \in D} t_u}{\sum_{u \in S_h, d_u \in D} \binom{d_u}{2}}.$$

We bound the overall error as follows:

$$|\widehat{WDCC}(D) - \mathrm{WDCC}(D)\,|$$
$$\leq \underbrace{|\mathrm{WDCC}(D) - Y(D)|}_{(i)\text{: node sampling error}} + \underbrace{|\widehat{WDCC}(D) - Y(D)|}_{(ii)\text{: triangle estimator error}}\,. \tag{5}$$

First, we bound term (i) of Eq. 5. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. For the upper bound, we have:

$$\mathbb{P}\left[Y(D) \geq \mathrm{WDCC}(D) + \eta\right]$$
$$= \mathbb{P}\left[\sum_{u,d_u \in D} X_u t_u \geq (\mathrm{WDCC}(D) + \eta) \sum_{u,d_u \in D} X_u \binom{d_u}{2}\right]$$
$$= \mathbb{P}\left[\sum_{u,d_u \in D} Z_u \geq 0\right],$$

by defining $Z_u = X_u \left(t_u - (\mathrm{WDCC}(D) + \eta)\binom{d_u}{2}\right)$. Note that Z_u's are independent. Let $Z = \sum_{u,d_u \in D} Z_u$. We have:

$$\mathbb{E}[Z] = p_h \sum_{u,d_u \in D} \left(t_u - \mathrm{WDCC}(D)\binom{d_u}{2} - \varepsilon\binom{d_u}{2}\right)$$
$$= -\varepsilon \sum_{u,d_u \in D} p_h \binom{d_u}{2} \geq -\varepsilon\, \mathrm{C}(D)\, p_h \binom{L}{2},$$

as $D = [L, U)$. Moreover, each $|Z_u|$ is upper bounded by $\binom{d_u}{2} \leq \binom{U}{2}$, and $\mathrm{Var}[Z] \leq p_h\, \mathrm{C}(D) \binom{U}{2}^2$. By Bernstein's inequality (Lemma 2.3):

$$\mathbb{P}[Z - \mathbb{E}[Z] \geq -\mathbb{E}[Z]] \leq 2\exp\left(\frac{-p_h^2 \varepsilon^2 \binom{L}{2}^2 (\mathrm{C}(D))^2}{2p_h\, \mathrm{C}(D) \binom{U}{2}^2 + \frac{2}{3}\mathrm{C}(D) \binom{U}{2}^2 p_h \varepsilon}\right),$$

which is $\leq 1/n^c$ for $p_h = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2\, \mathrm{C}(D)}\right)$, when $U = \Theta(L)$. An equivalent proof applies for the lower bound $Y(D) \leq \mathrm{WDCC}(D) - \eta$, after which we apply union bound over the two tail bounds.

Now, we bound term (ii) of Eq. 5, which controls the error of our triangle-estimator. The proof is equivalent to the one for Theorem 3.6 by defining variables accordingly. Thus:

$$\left|\widehat{WDCC}(D) - Y(D)\right| = \frac{1}{\sum_{u,d_u \in D}\binom{d_u}{2}} \left|\widehat{A}(D) - A(D)\right| \leq \varepsilon Y(D),$$

with high probability. Combining item (i), (ii) we proved, for $D = [L, U)$ with $L \leq U \leq \tau$, that:

$$\left|\widehat{WDCC}(D) - \mathrm{WDCC}(D)\right| \leq \eta + \varepsilon Y(D) \leq \eta + \varepsilon(\mathrm{WDCC}(D) + \eta)$$
$$= \varepsilon \cdot \mathrm{WDCC}(D) + (1+\varepsilon)\eta.$$

Result from the theorem follows then by rescaling η.

□

Next, we prove bounds for our tail estimator, which provides estimates of binned-degree wise clustering coefficient when, for $D = [L, U)$, we have $\tau \leq L$.

Theorem 3.8. *Consider degree interval $D = [L, U)$ such that $\tau \leq L < U$. For any $\varepsilon \in (0, 1/2)$, define $D^+ = [L(1-\varepsilon), U(1+\varepsilon)]$, and $D^- = [L(1+\varepsilon), U(1-\varepsilon)]$. For the quantity $\mathrm{NDCC}(D) = \frac{S_{LCC}(D)}{C(D)}$, we have, if*

$$\tau = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2 p_t}\right) \text{ and } p_{M_t} p_{A_t} = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2\, \mathrm{T}(D)}\right),$$

our tail estimator from Alg. ?? outputs a value $\widehat{NDCC}(D)$ such that:

$$\widehat{NDCC}(D) \in \left[(1-\varepsilon)\ \frac{S_{LCC}(D^-)}{\mathrm{C}(D^+)}, (1+\varepsilon)\ \frac{S_{LCC}(D^+)}{\mathrm{C}(D^-)}\right],$$

Proof. For the tail estimator, it holds (w.h.p.) $\widehat{d}_v \in (1 \pm \varepsilon) d_v$ by Lemma 3.1 item (iii), for $\tau = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2 p_t}\right)$. Our tail estimator from Alg. ?? computes estimates as:

$$\widehat{NDCC}(D) = \frac{\sum_{v \in S_t, \lceil \widehat{d}_v \rceil \in D} \frac{\widehat{t}_v}{\binom{\lceil \widehat{d}_v \rceil}{2}}}{\left|\left\{v \in S_t, \lceil \widehat{d}_v \rceil \in D\right\}\right|} \overset{(i)}{=} \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{\widehat{t}_v}{\binom{\lceil \widehat{d}_v \rceil}{2}}}{\left|\left\{v, \lceil \widehat{d}_v \rceil \in D\right\}\right|}, \quad (6)$$

where (i) follows from Lemma 3.1, item (iv) w.h.p. Define:

$$Y(D) = \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{t_v}{\binom{\lceil \widehat{d}_v \rceil}{2}}}{\left|\left\{v, \lceil \widehat{d}_v \rceil \in D\right\}\right|},$$

and $\widehat{X}(D)$ using true degrees (in the binomial coefficients):

$$\widehat{X}(D) = \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{t_v}{\binom{d_v}{2}}}{\left|\left\{v, \lceil \widehat{d}_v \rceil \in D\right\}\right|} = \frac{\widehat{S}_{LCC}(D)}{\widehat{C}(D)};.$$

From Corollary 3.3, for $\tau \geq 4$, we have:

$$Y(D) \in \frac{1}{1 \mp 3\varepsilon}\widehat{X}(D). \quad (7)$$

Define $X(D)$ using true degree over sum and count as:

$$X(D) = \frac{\sum_{v, d_v \in D} \frac{t_v}{\binom{d_v}{2}}}{|\{v, d_v \in D\}|} = \frac{S_{LCC}(D)}{C(D)};$$

Now, we have the same terms in $\widehat{X}(D), X(D)$, but ranging on a different domain due to slacked (approximated) degrees. To explicit the domain differences, we consider complementary cdf to leverage monotonicity; that is, define $\widehat{F}(d) = \sum_{v, \lceil \widehat{d}_v \rceil \geq d} t_v / \binom{d_v}{2}$ and $F(d) = \sum_{v, d_v \geq d} t_v / \binom{d_v}{2}$. We have $\widehat{S}_{LCC}([L, U)) = \widehat{F}(L) - \widehat{F}(U)$, and $S_{LCC}([L, U)) = F(L) - F(U)$. By Lemma 3.1, items (ii) and (iii) we have (w.h.p.):

$$\widehat{F}(d) \leq \sum_{v, d_v \geq \frac{d}{1+\varepsilon}} \frac{t_v}{\binom{d_v}{2}} \leq \sum_{v, d_v \geq (1-\varepsilon)d} \frac{t_v}{\binom{d_v}{2}} = F((1-\varepsilon)d),$$

and, similarly:

$$\widehat{F}(d) \geq \sum_{v, d_v \geq \frac{d}{1-\varepsilon}} \frac{t_v}{\binom{d_v}{2}} \geq \sum_{v, d_v \geq (1+\varepsilon)d} \frac{t_v}{\binom{d_v}{2}} = F((1+\varepsilon)d).$$

Consequently, for our estimator:

$$\widehat{S}_{LCC}([L, U)) = \widehat{F}(L) - \widehat{F}(U) \leq F((1-\varepsilon)L) - F((1+\varepsilon)U)$$
$$= S_{LCC}([(1-\varepsilon)L, (1+\varepsilon)U]),$$

and, similarly:

$$\widehat{S}_{LCC}([L, U)) = \widehat{F}(L) - \widehat{F}(U) \geq F((1+\varepsilon)L) - F((1-\varepsilon)U)$$
$$= S_{LCC}([(1+\varepsilon)L, (1-\varepsilon)U]).$$

By defining D^+, D^- as in the statement of the theorem, we have $\widehat{S}_{LCC}(D) \in [S_{LCC}(D^-), S_{LCC}(D^+)]$. With the same exact steps, we can derive $\widehat{C}(D) \in [C(D^-), C(D^+)]$. Combining with Eq. 7, we obtain:

$$Y(D) \in \frac{1}{1 \mp 3\varepsilon}\left[\frac{S_{LCC}(D^-)}{C(D^+)}, \frac{S_{LCC}(D^+)}{C(D^-)}\right].$$

Now, we bound $\widehat{NDCC}(D)$ with respect to $Y(D)$, which corresponds to the triangle-estimator error. This proof is equivalent to the one in Theorem 3.6. We just need to define w_Δ's accordingly and notice that $w_\Delta \in \left[\frac{1}{\binom{U(1+\varepsilon)}{2}}, \frac{3}{\binom{L(1-\varepsilon)}{2}}\right]$. Applying the same steps leads to $\widehat{NDCC}(D) \in (1 \pm \varepsilon)\ Y(D)$ w.h.p. by setting $p_M p_A = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2 T(D)}\right)$.

Notice that $T_t(D) = T(D)$ with high probability by Lemma 3.1 item (iv). Overall, we can write:

$$\widehat{NDCC}(D) \in \left[(1-\varepsilon)\ \frac{S_{LCC}(D^-)}{\mathrm{C}(D^+)}, (1+\varepsilon)\ \frac{S_{LCC}(D^+)}{\mathrm{C}(D^-)}\right],$$

as $\varepsilon \in (0, 1/2)$, and by rescaling, concluding the proof. □

Theorem 3.9. *Consider degree interval $D = [L, U)$ such that $\tau \le L < U$. For any $\varepsilon \in (0, 1/2)$, define $D^+ = [L(1-\varepsilon), U(1+\varepsilon)]$, and $D^- = [L(1+\varepsilon), U(1-\varepsilon)]$. For the quantity $\mathrm{WDCC}(D) = \frac{\mathrm{T}(D)}{\mathrm{W}(D)}$, we have, if*

$$\tau = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2 p_t}\right) \text{ and } p_{M_t} p_{A_t} = \widetilde{\Omega}\left(\frac{\Delta_E}{\varepsilon^2\, \mathrm{T}(D)}\right),$$

our tail estimator from Alg. ?? outputs a value $\widehat{WDCC}(D)$ such that:

$$\widehat{WDCC}(D) \in \left[(1-\varepsilon)\ \frac{\mathrm{T}(D^-)}{\mathrm{W}(D^+)}, (1+\varepsilon)\ \frac{\mathrm{T}(D^+)}{\mathrm{W}(D^-)}\right],$$

Proof. The proof follows equivalent steps of the proof of Thm. 3.8, by applying complementary cdf on $\mathrm{T}(D)$ and on $\mathrm{W}(D)$. □

Now, we prove our result on local clustering coefficient for sufficiently high-degree nodes (i.e., from our tail estimator).

Theorem ?? from main paper. *Let $v \in S_t$ be a node such that $d_v \ge \tau$ and $LCC(v) > 0$. For any $\varepsilon \in (0, 1/2)$ and any $\eta \in (0, LCC(v)]$, if*

$$\tau = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2 p_t}\right), \text{ and } p_{M_t} p_{A_t} = \widetilde{\Omega}\left(\frac{\Delta_E(v) LCC(v)}{\eta^2 \tau^2}\right),$$

then, with high probability,

$$\left|\widehat{LCC}(v) - LCC(v)\right| \le \varepsilon LCC(v) + \eta.$$

Proof. First, a node v with degree $d_v \ge \tau$ is included in S_t for the above choice of τ with high probability, from Lemma 3.1. Our tail estimator computes, for any such node:

$$\widehat{LCC}(v) = \frac{\widehat{t}_v}{\binom{\lceil \widehat{d}_v \rceil}{2}} \overset{(i)}{\in} \frac{1}{1 \mp \varepsilon} \frac{t_v}{\binom{d_v}{2}} := \frac{1}{1 \mp \varepsilon} Y(v),$$

where (i) follows from Corollary 3.3, and by rescaling ε. For vertex v, let $X_i(v) = 1/p_{\Delta_i}$ if i-th triangle is incident to v and is counted by Alg. ??. We have $\mathbb{E}[X_i(v)] = 1$, so that $\widehat{t}_v = \sum_i X_i(v)$ and $\mathbb{E}\left[\widehat{t}_v\right] = t_v$. Moreover, $\mathrm{Var}[X_i(v)] \le 1/p_{\Delta_i} \le 1/p_A p_M$, assuming $p_A \le p_M$. We can build a dependency graph for variables $X_i(v)$ for all i, with fractional chromatic number $\le \Delta_E(v)$, which is the maximum number of triangles incident to an edge that contains node v. Define centered variables $Z_i(v) = X_i(v) - \mathbb{E}[X_i(v)]$, and $Z = \sum_i Z_i(v)$. By Lemma 2.5:

$$\mathbb{P}\left[|Y(v) - LCC(v)| \ge \eta\right] = \mathbb{P}\left[|Z| \ge \eta \binom{d_v}{2}\right]$$
$$\le 2\exp\left(\frac{-8\eta^2 \binom{d_v}{2}^2}{25\Delta_E(v)\left(\frac{t_v}{p_A p_M} + \frac{\eta\binom{d_v}{2}}{3 p_A p_M}\right)}\right),$$

which is $\le 1/n^c$ for $p_A p_M = \widetilde{\Omega}\left(\frac{\Delta_E(v) LCC(v)}{\eta^2 \tau^2}\right)$ for $\eta \le LCC(v)$, and since $d_v \ge \tau$. Overall:

$$\left|\widehat{LCC}(v) - LCC(v)\right| \le \left|\widehat{LCC}(v) - Y(v)\right| + |Y(v) - LCC(v)|$$
$$\le \varepsilon Y(v) + \eta \le \varepsilon LCC(v) + \eta(\varepsilon + 1).$$

Main result follows by rescaling ε and η.

□

Now, we report the proof for the equivalence of tail triangle estimator in the two-pass and in the one-pass algorithms.

Claim 3.10. *Assuming $p_{M_t} = p_{A_t}$, for every node $v \in S_t$ the number of triangles incident to v counted by the one-pass algorithm is identical to the number counted by the two-pass algorithm. Consequently, the tail estimator produced by the one-pass algorithm coincides with the one produced by the two-pass algorithm on S_t.*

Proof. We prove the claim by coupling the randomness of the one-pass and two-pass algorithms. For each edge e in the stream, independently pre-generate a Bernoulli random variable X_e with parameter p_t, used to determine whether $e \in E_t$, and a Bernoulli random variable Y_e with parameter $p_{M_t} = p_{A_t}$, used to determine whether e is sampled for tail triangle counting. Consider the same variables $\{X_e, Y_e\}_e$ for both algorithms.

Since both algorithms use the same Bernoulli variable X_e to decide whether an edge e belongs to E_t, the resulting set E_t is identical. As S_t is determined solely by E_t, the set S_t is also identical in the two algorithms.

Now, fix any node $v \in S_t$, and consider any triangle of the type $\Delta = \{\{v, a\}, \{v, b\}, \{a, b\}\}$ incident to v. Let e_1, e_2 be the two edges of Δ that appear before the closing edge in the stream.

We first analyze the two-pass algorithm. After the first pass, the set S_t is fixed. An edge is inserted into the tail main sample if it is incident to a node of S_t and $Y_e = 1$; otherwise, if $Y_e = 1$, it is inserted into the tail auxiliary sample. Since $v \in S_t$, at least one of the two edges e_1, e_2 is incident to v. Hence, whenever that edge is sampled, it must belong to the tail main sample. Therefore, the event that Δ is counted by the two-pass algorithm is exactly $Y_{e_1} = 1$ and $Y_{e_2} = 1$.

Consider now the one-pass algorithm. At the arrival time of an edge e, the algorithm decides, based on the current stream prefix, whether e is eligible for the tail main sample or for the tail auxiliary sample. Since $p_{M_t} = p_{A_t}$, in both cases the decision is made using the same variable Y_e. Thus, independently of whether e is currently being sampled in the main or auxiliary sample, the event that e is sampled is $Y_e = 1$. The one-pass algorithm counts a triangle whenever its two non-closing edges are sampled, even if both are stored in the auxiliary sample. Hence, for the same triangle Δ, the event that it is counted is again exactly $Y_{e_1} = 1$ and $Y_{e_2} = 1$.

Thus, for every triangle incident to $v \in S_t$, the counting event is identical in the one-pass and two-pass algorithms. Summing over all such triangles shows that the number of counted triangles incident to v is the same in the two algorithms, for every $v \in S_t$. □

Finally, we report the proof about space and time complexity of our proposed one-pass algorithm.

Theorem ??. *Let $\mathcal{D} = \{D_1, D_2, \dots\}$ be a set of degree intervals. There exists a* one-pass *streaming algorithm that, for all $D \in \mathcal{D}$* simultaneously, *computes approximations of* $\mathrm{NDCC}(D)$ *and of* $\mathrm{WDCC}(D)$ *with the guarantees of Theorems* ?? *and* ??, *using* expected *space:*

$$\widetilde{O}\left(\frac{n}{\eta^2} \max_{D=[L,U)\in\mathcal{D}:\, U\le\tau} \left(\frac{1}{\mathrm{C}(D)} + \frac{\Delta_V(D)}{\mathrm{T}(D)}\right)\right) \text{ nodes,}$$

and

$$\widetilde{O}\left(m\left(p_t + \frac{\eta}{\varepsilon} \max_{D\in\mathcal{D}} \sqrt{\frac{\Delta_E(D)}{\Delta_V(D)}}\right)\right) \text{ edges,}$$

where τ is the degree threshold selected by Algorithm ??.

Proof. The overall pipeline is as follows: first, run our one-pass algorithm to gather node-degree and node-triangle estimates. Parameters of the one-pass algorithm should be fixed accordingly to $\mathcal{D}$: if all $D = [L, U) \in \mathcal{D}$ are such that $\tau \ge L$, then all intervals are estimated by using the tail estimator (i.e., there is no point in using the head estimator and hence one can set $p_h = 0$). In general, p_h should consider only degrees $D = [L, U) \in \mathcal{D}$: $U \le \tau$. Note that $\tau = c/(\varepsilon^2 p_t)$ can be computed upfront, given our independent parameter p_t. Upon running the one-pass algorithm, for every degree interval $D = [L, U) \in \mathcal{D}$ run Algorithm ?? to obtain estimates $\widehat{NDCC}(D)$ and $\widehat{WDCC}(D)$, either using head estimator (if $U \le \tau$) or the tail estimator (if $\tau \ge L$)).

We plug bounds derived in Thm. 3.6, 3.7, 3.8, 3.9. The space used by the algorithm accounts for the node-degree map S_h (head), the edge sample E_t (tail), and the subgraph stored for counting triangles. The expected number of nodes in S_h is $n \cdot p_h$, and the expected number of edges in the sample E_t is $m \cdot p_t$. The expected number of edges for counting triangles is bounded by $m\left(p_{M_h} + p_{A_h} + p_{M_t} + p_{A_t}\right)$, subject to constraints $p_{M_h} p_{A_h} = \widetilde{\Omega}\left(\frac{\Delta_E(D)}{\varepsilon^2 p_h\, \mathrm{T}(D)}\right)$, and $p_{M_t} p_{A_t} = \widetilde{\Omega}\left(\frac{\Delta_E(D)}{\varepsilon^2\, \mathrm{T}(D)}\right)$, where $\Delta_E(D)$ is the maximum number of triangles incident to an edge, with such edge containing a node with degree in D. Thus, the quantity $p_{M_h} + p_{A_h} + p_{M_t} + p_{A_t}$ is minimised when $p_{M_h} = p_{A_h} = \widetilde{O}\left(\sqrt{\frac{\Delta_E(D)}{\varepsilon^2 p_h\, \mathrm{T}(D)}}\right)$ and $p_{M_t} = p_{A_t} = \widetilde{O}\left(\sqrt{\frac{\Delta_E}{\varepsilon^2\, \mathrm{T}(D)}}\right)$. We have $p_h = \widetilde{\Omega}\left(\frac{1}{\eta^2}\left(\frac{1}{\mathrm{C}(D)} + \frac{\Delta_V(D)}{\mathrm{T}(D)}\right)\right)$, and $\frac{\mathrm{T}(D)}{\mathrm{C}(D)} \le \Delta_V(D)$. Plugging and re-arraging, yields the final bound for a single degree interval $D \in \mathcal{D}$. Main result follows by a union bound on $|\mathcal{D}| = O(n^2)$ and considering the interval D which maximizes each quantity depending on it. □

Proposition ??. *Given an input stream Σ for graph G with m edges and maximum degree $d_{max}(G)$, our one-pass algorithm processes the stream in $O(m \cdot d_{max}(G))$ time.*

Proof. For each incoming edge $\{u, v\}$ in the stream Σ, the most expensive step is the computation of common neighbors when counting triangles. For edge $\{u, v\}$, computing common neighbors takes $O(\min(d_u, d_v))$, where adjacency lists of subgraph are implemented by hash sets, so we can iterate over the smaller set and use set membership in the largest one. Iterating over all the edges, it trivially follows $\sum_{\{u,v\}\in\Sigma} O(\min(d_u, d_v)) = O(m \cdot d_{max}(G))$, where $d_{max}(G)$ is the maximum degree for graph G, and is a rough upper bound as we need to consider degrees in the subgraph induced by sampled edges. All remaining operations, including hash computations, accesses to the node-degree map S_h, and insertions into E_t, M_h, M_t, A_h, and A_t, take constant time per operation under standard hashing assumptions. The final scan used to compute tail nodes takes at most $O(m)$ time, since $|E_t| \le m$, and is therefore absorbed by $O(m \cdot d_{\max}(G))$. □

4 BOLIDE ALGORITHM

In this section, we detail our one-pass algorithm BOLIDE, implemented and used in our experiments. As discussed in Section ??, the sampling of nodes, tracking of degrees and choice of threshold τ in implemented as in headtail [7], while the counting of triangles uses reservoir sampling to keep the memory fixed with certainty, in place of fixed probability sampling.

Algorithm 1: BOLIDE$(\Sigma, p_h, p_t, B_{M_h}, B_{M_t}, B_{A_h}, B_{A_t})$

Input : Graph stream Σ;
Probabilities p_h, p_t of sampling head and tail nodes;
Budgets B_{M_h}, B_{M_t} for head and tail main samples;
Budgets B_{A_h}, B_{A_t} head and tail auxiliary samples.

Output : Estimate of degrees and estimates of triangles for head and tail nodes; degree threshold.

1 Init head and tail node-degree maps $S_h, S_t \longleftarrow \emptyset$;
2 Init head and tail node-triangle maps: $\hat{T}_h, \hat{T}_t \longleftarrow \emptyset$;
3 Init reservoir of edges $M_h, M_t, A_h, A_t \longleftarrow \emptyset$;
4 Init number h, t of head and tail edges: $h, t \longleftarrow 0$;
5 Init number a_h, a_t of aux head and aux tail edges: $a_h, a_t \longleftarrow 0$;
6 **for** each incoming edge $e = \{u, v\}$ in the stream Σ **do**
7   **if** $u \notin S_h$ **then**
8   **if** $hash[u] < p_h$ **then** $S_h[u] \longleftarrow 1$;
9   **else** $S_h[u] \longleftarrow S_h[u] + 1$;
10   **if** $u \notin S_t$ **then** Set $S_t[u] \longleftarrow 1$ with probability p_t;
11   **else** $S_t[u] \longleftarrow S_t[u] + 1$;
12   Repeat the above steps 7-11 also for v;
13   CTReservoir$(\{u, v\}, M_h, B_{M_h}, h, A_h, B_{A_h}, a_h, \hat{T})$;
14   CTReservoir$(\{u, v\}, M_t, B_{M_t}, t, A_t, B_{A_t}, a_t, \hat{T})$;
15   **if** $u \in S_h$ *or* $v \in S_h$ **then** SEReservoir$(\{u, v\}, M_h, B_{M_h}, h)$;
16   **else** SEReservoir$(\{u, v\}, A_h, B_{A_h}, a_h)$;
17   **if** $u \in S_t$ *or* $v \in S_t$ **then** SEReservoir$(\{u, v\}, M_t, B_{M_t}, t)$;
18   **else** SEReservoir$(\{u, v\}, A_t, B_{A_t}, a_t)$;
19 Let $\ell(r) = \lceil \frac{1-p_t-(1-p_t)^{r+1}-rp_t(1-p_t)^r}{p_t(1-(1-p_t)^r)} \rceil$;
20 Set $S_t[u] \longleftarrow (S_t[u] - \ell(S_t[u]))$ for all $u \in S_t$;
21 Let $C_h(d)$ be the number of nodes with degree exactly d in S_h;
22 Set τ to be the largest degree d such that $C_h(d)/p_h \ge 10/p_h$;
23 **return** head and tail node-degree estimates S_h, S_t, head and tail triangle estimates $\hat{T}_h, \hat{T}_t$, threshold τ.

The pseudocode of our algorithm BOLIDE is reported in Alg. 1, and described in the following. Again, we assume the existence of a hash function $hash : V \longrightarrow [0, 1]$ that maps nodes uniformly to the interval $[0, 1]$. First, we initialize the node-degree maps S_h, S_t, head and tail main samples M_h, M_t, head and tail auxiliary samples A_h, A_t, node-triangles maps $\hat{T}_h, \hat{T}_t$, number h, t, a_h, a_t for tracking edges in each respective set to compute reservoir sampling probabilities (lines 1-5). For each incoming edge $e = \{u, v\}$ in the stream (line 6), the algorithm performs the following steps. Node sampling

for head nodes is performed as described in Alg. **??** That is, if the incoming node $u \notin S_h$, then u is added to S_h if $hash[u] < p_h$, and its degree is initialized. Otherwise, if $u \in S_h$ its degree is incremented (lines 7-9). Node sampling for tail nodes is performed with sample-and-hold strategy (see Sect. **??**): if $u \notin S_t$, then u is added to S_t with independently with probability p_t, and its degree is initialized. Otherwise, if $u \in S_t$, its degree is incremented (lines 10, 11). The above steps are then repeated also for node v (line 12). Next, Alg. 1 employs triangle estimators. That is, triangles closed by e in main and auxiliary sample are counted, for triangles incident to head nodes (line 13) and to tail nodes (line 14). The incoming edge is then sampled into the reservoir, depending on whether it belongs to S_h (head main sample, line 15) or not (head auxiliary sample, line 16), and to S_t (tail main sample, line 17) or not (tail auxiliary sample, line 18). At the end of the stream, tail degrees are corrected as explained in [7] (lines 19, 20), and threshold τ is computed (as proposed in the implementation of [7]) in lines 21, 22. Finally, head node-degrees, head node-triangles, tail node-degrees, tail node-triangles maps, and threshold τ are returned (line 23).

Algorithm 2: `CTReservoir`$(\{u,v\}, S, M_s, s, A, M_a, a, \hat{T})$

Input : edge $\{u,v\}$;
main sample S; memory budget M_s for sample S;
number s of edges belonging to S;
auxiliary sample A; memory budget M_a for sample A;
number a of edges belonging to A;
Node-triangle map $\hat{T}$.

Let $Adj_S(u), Adj_S(v)$ be the adjacency lists of u, v in sample S;
Let $Adj_A(u), Adj_A(v)$ be the adjacency lists of u, v in sample A;
Let $N(u) \longleftarrow Adj_S(u) \cup Adj_A(u)$;
Let $N(v) \longleftarrow Adj_S(v) \cup Adj_A(v)$;
for *each node* $w \in N(u) \cap N(v)$ **do**
  Initialize $\hat{T}[u], \hat{T}[v], \hat{T}[w]$ to zero if not set yet;
  if $w \in Adj_S(u)$ AND $w \in Adj_S(v)$ **then**
    $p \longleftarrow \min\left(1, \frac{M_s}{s} \cdot \frac{M_s - 1}{s-1}\right)$;
  else if $w \in Adj_A(u)$ AND $w \in Adj_A(v)$ **then**
    $p \longleftarrow \min\left(1, \frac{M_a}{a} \cdot \frac{M_a - 1}{a-1}\right)$;
  else $p \longleftarrow \min\left(1, \frac{M_s}{s} \cdot \frac{M_a}{a}\right)$;
  Increment $\hat{T}[u], \hat{T}[v], \hat{T}[w]$ by p;

Algorithm 3: `SEReservoir`$(\{u,v\}, S, M, s)$

Input : edge $\{u,v\}$; edge sample S;
memory budget M for sample S;
number s of edges belonging to S seen so far.

if $|S| < M$ **then** $S \longleftarrow S \cup \{\{u,v\}\}$;
else if `FlipBiasedCoin`$(M/s) == HEAD$ **then**
  $\{\bar{u}, \bar{v}\} \longleftarrow$ edge sampled uniformly at random from S;
  $S \longleftarrow (S \setminus \{\{\bar{u}, \bar{v}\}\}) \cup \{\{u,v\}\}$;
$s \longleftarrow s + 1$;

Algorithm 1 makes use of two subroutines: `CTReservoir` and `SEReservoir`. `CTReservoir` (Algorithm 2) is given the incoming edge $\{u,v\}$, the main sample S with its memory budget M_s and counter s, the auxiliary sample A with its budget M_a and counter a, and the node-triangle map to eventually update. First, the algorithm computes the adjacency lists of nodes u, v in both samples S and A, deriving the list of neighbors (lines 1-4). Then, for every node w forming a triangle with u and v (line 5), it derives the probability of sampling such triangle based on whether the two edges (when the incoming one arrives) forming the triangle belong both to sample S (line 8), both to sample A (line 10), or one in S and the other in A (line 11). At the end, such probability is used to increment the local triangles count (line 12). `SEReservoir` (Algorithm 3) is given the edge to sample, the reservoir sample S together with its memory budget M and the number s of elements seen so far (belonging to sample S), and applies the reservoir sampling update: if sample S is not full yet, then it adds the incoming edge to S (line 1), otherwise it adds it to the sample with probability M/s and by replacing an edge in S uniformly at random (lines 2-4). At the end, the algorithm increments the counter s (line 5). We assume that `FlipBiasedCoin`(p) returns $HEAD$ with probability p.

Notice that correctness of our triangle-counter algorithm follows from the use of reservoir sampling (see [2, 5, 6, 8, 9]), and from the fact we are using independent reservoir samples for (head and tail) main and auxiliary samples.

5 ADDITIONAL EXPERIMENTS

In this section, we first assess empirically the quality of bounds obtained by our tail estimator. Moreover, we analyze the extra number of counters necessary for guaranteeing correct estimates for our final one-pass algorithm. Then, we motivate the choice of parameters for our algorithm `BOLIDE`, together with analyzing the impact of the size of the auxiliary sample. Finally, we report results missing in the main paper due to space issues.

5.1 Empirical Bounds for Tail Estimator

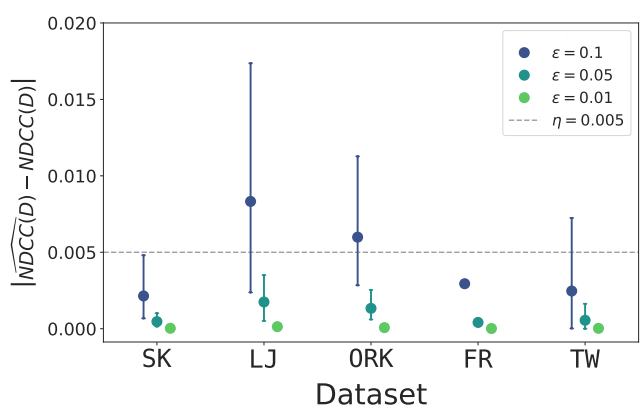

Figure 1: Absolute error of estimates given by bound in Eq. 8, for various values of ε. For each dataset and ε, average and standard deviation is reported across 20 different intervals $D = [2^i, 2^{i+1})$, such that $\tau \leq 2^i$. Dashed line indicates $\eta = 0.005$, the absolute floor noise in our experiments.

In the following, we empirically assess the quality of the bounds reported by our tail estimator. For degree interval D, let $\text{NDCC}(D) =$

$S_{LCC}(D)/\mathrm{C}(D)$. From Theorem **??**, our algorithm ouputs:

$$\widehat{NDCC}(D) \in \left[(1-\varepsilon)\frac{S_{LCC}(D^-)}{\mathrm{C}(D^+)}, (1+\varepsilon)\frac{S_{LCC}(D^+)}{\mathrm{C}(D^-)}\right], \quad (8)$$

for any degree interval $D = [L, U)$ such that $L \geq \tau$, where $D^- = [(1+\varepsilon)L, (1-\varepsilon)U]$ and $D^+ = [(1-\varepsilon)L, (1+\varepsilon)U]$.

To empirically quantify such bounds, we compute absolute error $|\widehat{NDCC}(D) - \mathrm{NDCC}(D)|$ considering degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1})$). We consider values of $\widehat{NDCC}(D)$ as the middle point with respect to the interval guarantees from Equation 8. We pick 20 intervals equally spaced, making sure that for each interval $D = [L, U)$ we have $L \geq \tau$ (i.e., when we employ our tail estimator), for values of τ computed by our algorithm BOLIDE in the experiments (see Table **??** for values of τ).

Figure 1 shows results for $\varepsilon \in \{0.01, 0.05, 0.1\}$. While for $\varepsilon = 0.1$ we observe large absolute errors and high variance, lower values of ε (i.e., tighter intervals D^-, D^+) allow for very accurate and precise stable bounds, across all the datasets and the considered intervals. This means that in real-world networks, the distribution of degree-binned clustering coefficient is stable when allowing small slack in the binned intervals (for high degrees), and our bounds in Theorem **??** are empirically accurate.

5.2 One Pass Algorithm: Extra Counters

In this section, we investigate the number of extra counters needed for ensuring unbiasedness of estimates for our one-pass algorithm. Let N be the number of nodes for which we know the degree, and N' be the number of nodes for which we know the number of local triangles (i.e., triangle incident to such nodes). In our two-pass algorithm, we have $N \geq N'$, given that our second pass SecondPass (Alg. **??**) is estimating triangles only incident to nodes for which we know the degree from the first pass FirstPass (Alg. **??**). The number of the former and latter nodes is precisely N', N respectively. Conversely, when we consider our one-pass algorithm, we also count triangles incident to nodes for which (currently) we are unaware of their degrees. Such nodes belong to edges in the auxiliary sample; counting such triangles allows for the unbiasedness (see Sect. **??**). That is, for our one-pass algorithm we may have $N \leq N'$. Note that the extra counters for triangles will be wasted, as we cannot provide an estimate of clustering coefficients without knowing degrees (or estimates of degrees) for such nodes. Table 1 shows the fraction of extra counters stored by our one-pass algorithm. We compute $c = N'/N$ for each dataset, and for different size of auxiliary sample (i.e., $M_{A_h} + M_{A_t} = f \cdot m$, with $M_{A_h} = M_{A_t}$, and for values of $f \in \{0.01, 0.05, 0.1\}$, showed on the table.) We observe that, in practice, the number of extra counters is within a small constant than the number of counters needed. Over all the datasets, we only need to store from 2.73 and up to 8.22 times the number of counters effectively needed.

5.3 Empirical Values $\Delta_V(D)$ vs $\Delta_E(D)$

In the following, we inspect values of $\Delta_V(D), \Delta_E(D)$ in real-world datasets. Recall that, for a degree interval D, $\Delta_V(D)$ is the maximum number of triangles incident to a node having degree in D, and $\Delta_E(D)$ is the maximum number of triangles incident to an edge, with such edge containing a node with degree in D. The space complexity of our one-pass algorithm depends on $\Delta_E(D)/\Delta_V(D)$ (see Theorem **??**). Trivially, it follows that $\Delta_V(D) \geq \Delta_E(D)$. Table 2 shows values of $\mathrm{C}(D)$ and $\mathrm{T}(D)$, together with $\Delta_E(D), \Delta_V(D)$, and ratio $\Delta_E(D)/\Delta_V(D)$, for all the intervals of power of two (i.e., $D = [2^i, 2^{i+1})$) considering $i \geq 2$, for SK, LJ and ORK. We see that, in practice, the ratio $\Delta_E(D)/\Delta_V(D)$ is consistently and significantly small, with $\Delta_V(D)$ exceeding $\Delta_E(D)$ by up to 3 orders of magnitude, confirming that the space used by our algorithm is very low.

Table 1: Number $c = N'/N$ of extra counters stored by our one-pass algorithm, where N, N' are the number of nodes for which we estimate their degree, and the number of nodes for which we estimate triangles. For each dataset and choice of auxiliary sample budget, mean and standard deviation over 10 independent repetitions are reported.

	Auxiliary Samples Budget		
Dataset	$0.01m$	$0.05m$	$0.10m$
SK	5.70 ± 0.19	6.99 ± 0.09	8.22 ± 0.04
LJ	2.73 ± 0.01	4.11 ± 0.01	5.19 ± 0.01
ORK	4.70 ± 0.01	6.69 ± 0.01	7.80 ± 0.01
TW	4.88 ± 0.03	6.05 ± 0.02	7.06 ± 0.02
FR	3.33 ± 0.00	4.36 ± 0.00	5.14 ± 0.00

5.4 Choice of Parameters

In the following, we explain the choice of parameters for our algorithm BOLIDE. Given that we have many parameters, and it for each parameter it yields the higher the better (as they are function of the space used), we first constrain BOLIDE to use exactly $11\%m$ edges: we consider pairs $(M_h, M_t) = \{(0.01, 0.09), (0.05, 0.05), (0.09, 0.01)\} \cdot m$ to fix the size of our reservoir samples, and we set $M_{A_h} = M_{A_t} = 0.005 \cdot m$ for the auxiliary samples. In this way, note that BOLIDE uses precisely at $11\%m$ edges, at any time in the stream [9]. We then consider values of $p_h \in \{0.1, 0.2, 0.3\}$ and values of $p_t \in \{0.05, 0.1, 0.15\} \cdot n/m$ and we perform a grid search to identify the best choice of parameters.

Given that these experiments do not involve scalability of our algorithm, we do not consider our biggest datasets (TW and FR). Figure 2 shows results. In the x-axis we consider values of p_t, and in the y-axis we report RHAS for distribution $\{NDCC(D_i)\}_i$ and degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1})$). For each row (dataset), a subplot represents a different value of p_h. We see that the choice $M_h = 0.09m, M_t = 0.01m$ is consistently the best and the most stable, but for SK and ORK only with $p_h = 0.1$, where it is the second best. We see the opposite assignment (i.e., $M_h = 0.01m, M_t = 0.09m$) is performing badly and exhibits high variance. The reason is that, in real-world datasets, degree distribution follows a heavy-tailed shape, and there are many more edges incident to low-degree nodes (as they are a lot) conversely to high-degree nodes (as they are few). Moreover, for the choice $M_h = 0.09m, M_t = 0.01m$, we observe a discrete improvement on RHAS when going from $p_h = 0.1$ to $p_h = 0.2$, and a negligible one when going from $p_h = 0.2$ to $p_h = 0.3$, and stable performance across values of p_t. As a result, we pick $p_h = 0.2, p_t = 0.1 \cdot n/m$, so that we have (in expectation) the number

Table 2: Table reporting, for each degree interval $D = [2^i, 2^{i+1})$ with $i \geq 2$, the quantities $\mathrm{C}(D)$ (number of nodes with degree in D), $\mathrm{T}(D)$ (number of triangles incident to nodes with degree in D), $\Delta_V(D)$ (maximum number of triangles incident to a node with degree in D), $\Delta_E(D)$ (maximum number of triangles incident to an edge, with such edge containing a node with degree in D), and ratio $\Delta_E(D)/\Delta_V(D)$.

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$[2^i, 2^{i+1})$	$\mathrm{C}(D)$	$\mathrm{T}(D)$	$\Delta_V(D)$	$\Delta_E(D)$	$\frac{\Delta_E(D)}{\Delta_V(D)}$	$\mathrm{C}(D)$	$\mathrm{T}(D)$	$\Delta_V(D)$	$\Delta_E(D)$	$\frac{\Delta_E(D)}{\Delta_V(D)}$	$\mathrm{C}(D)$	$\mathrm{T}(D)$	$\Delta_V(D)$	$\Delta_E(D)$	$\frac{\Delta_E(D)}{\Delta_V(D)}$
$[2^2, 2^3)$	456k	1M	21	6	0.2857	834k	3M	21	6	0.2857	148k	595k	21	6	0.2857
$[2^3, 2^4)$	319k	4M	105	14	0.1333	743k	11M	105	14	0.1333	300k	4M	105	14	0.1333
$[2^4, 2^5)$	150k	7M	429	30	0.0699	617k	28M	465	30	0.0645	555k	25M	433	30	0.0693
$[2^5, 2^6)$	69k	9M	1.48k	62	0.042	404k	52M	1.95k	62	0.0317	772k	100M	1.61k	62	0.0385
$[2^6, 2^7)$	22k	7M	4.26k	123	0.0289	185k	71M	7.87k	126	0.016	685k	247M	5.34k	124	0.0232
$[2^7, 2^8)$	9.36k	8M	10k	246	0.0244	62k	110M	32k	254	0.008	333k	326M	16k	240	0.015
$[2^8, 2^9)$	3.71k	8M	18k	474	0.0259	19k	158M	114k	510	0.0045	96k	239M	30k	420	0.0139
$[2^9, 2^{10})$	1.4k	9M	34k	999	0.0292	4.86k	92M	214k	995	0.0047	30k	199M	58k	850	0.0147
$[2^{10}, 2^{11})$	536	6M	69k	2.02k	0.0292	704	18M	229k	1.39k	0.0061	3k	75M	118k	1.73k	0.0147
$[2^{11}, 2^{12})$	211	5M	107k	3.56k	0.0333	161	5M	123k	2.54k	0.0207	735	49M	266k	3.13k	0.0118
$[2^{12}, 2^{13})$	85	3M	154k	7.62k	0.0495	29	2M	123k	4.3k	0.035	250	35M	591k	5.85k	0.0099
$[2^{13}, 2^{14})$	60	4M	283k	16k	0.0558	11	1M	368k	4.45k	0.0121	99	28M	1M	9.14k	0.009
$[2^{14}, 2^{15})$	24	4M	531k	29k	0.0539	1	59k	59k	840	0.0142	27	17M	2M	8.55k	0.0051
$[2^{15}, 2^{16})$	3	1M	565k	29k	0.0508	–	–	–	–	–	1	708k	708k	9.07k	0.0128

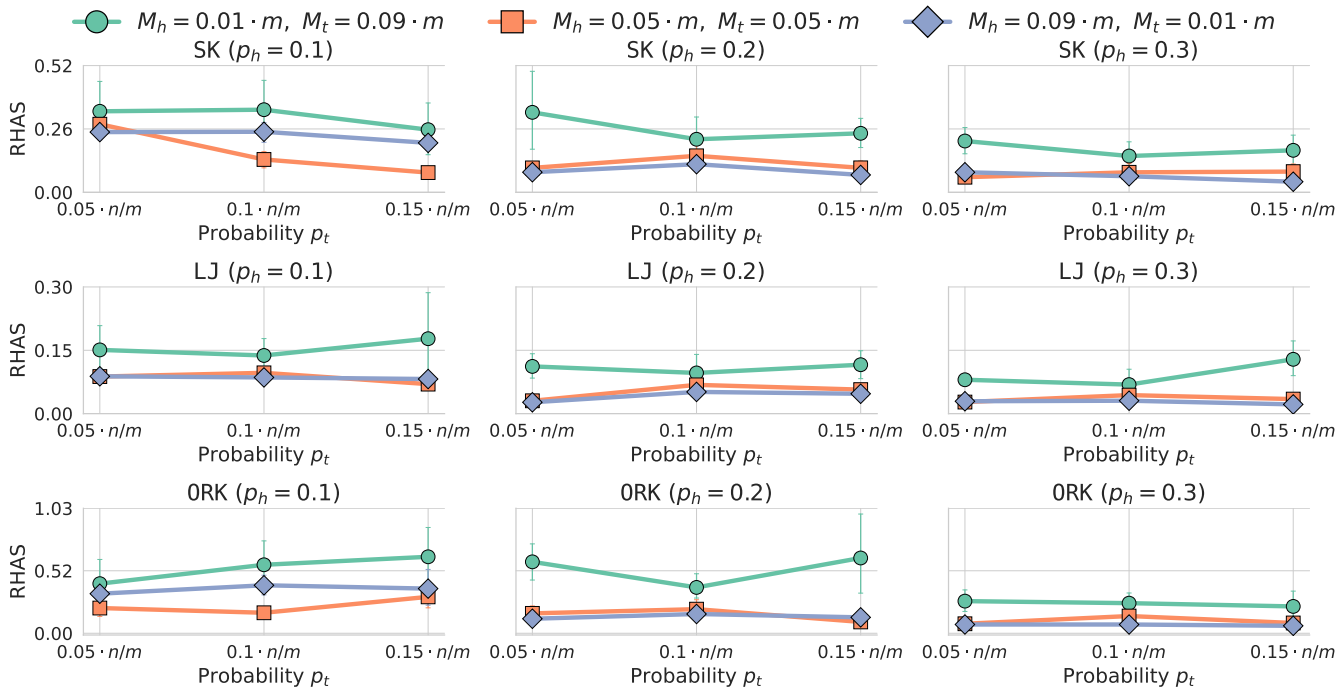

Figure 2: RHAS distance relative to NDCC distribution for degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1}]$) versus probability p_t of sampling tail nodes. For each row (dataset), each subplot shows a different value of probability p_h of sampling head nodes. For each dataset and choice of parameters, average and 90% confidence intervals over 10 independent repetitions are shown. Y-axis is shared across rows (datasets).

of nodes in our samples to be $0.3n$ (excluding duplicates, which can be easily ignored).

5.5 Impact of Auxiliary Sample

Next, we assess the accuracy of estimates when varying the size of our auxiliary sample. We run BOLIDE with parameters fixed as

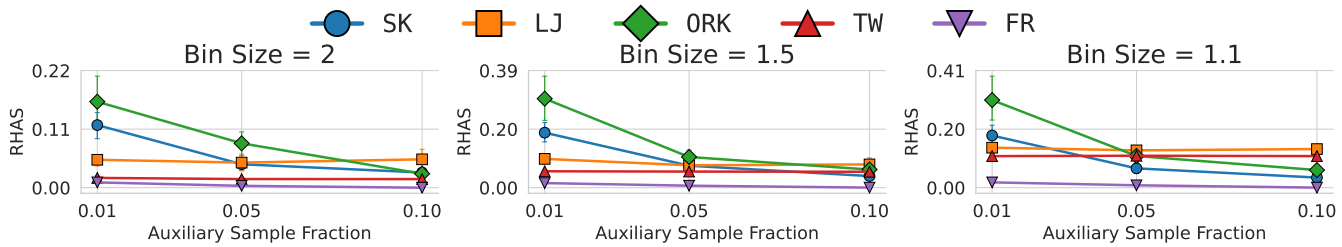

Figure 3: RHAS distance relative to NDCC distribution for degree intervals of powers of $\{2, 1.5, 1.1\}$ (subplots) versus auxiliary samples size, as a fraction of m. For each dataset and choice of parameters, average and 90% confidence intervals over 10 independent repetitions are shown.

detailed in Section 5.4. Then, for head and tail auxiliary samples we consider values $M_{A_h} + M_{A_t} = f \cdot m$, with $M_{A_h} = M_{A_t}$, and for values of $f \in \{0.01, 0.05, 0.1\}$, showed on the x-axis. That is, we assign 1%, 5% and 10% of the number m of edges to our auxiliary samples, equally distributed between head and tail. Fig. 3 shows the RHAS distance relative to NDCC computed on degree intervals of powers of 2, 1.5 and 1.1, reported on each subplot. We see that increasing the auxiliary brings decent improvements for SK and ORK, while is negligible for the others, as BOLIDE provides accurate estimates even with $f = 0.01$.

5.6 Additional Results

In the following, we report results missing from Section **??** due to space issues. That is, we display estimates of our algorithm BOLIDE for $\{NDCC(D_i)\}_i$ and $\{WDCC(D_i)\}_i$ distributions, considering degree intervals $D_i = [b^i, b^{i+1})$, for $b = \{2, 1.5, 1.2, 1.1\}$. Figures 4, 5 depict results for NDCC for $b = \{2, 1.1\}$, while Figures 6 and 7 show results for WDCC for $b = \{2, 1.5, 1.2, 1.1\}$. Similarly to what observed in Section **??**, we see that BOLIDE is qualitatively good, as the estimated distribution of NDCC and WDCC is very close to the true distribution. Moreover, our algorithm is quantitatively accurate given that RHAS distance is consistently well below 15%, execpt for the low-degree nodes of SK and ORK. For the latter datasets, allowing for more budget for auxiliary samples helps in increasing accuracy, as highlighted in Section 5.5.

Comparison with Wedge-Sampling. In Table 3 we list the parameters used to enforce a similar RHAS (with tolerance 2%) between BOLIDE and Wedge-Sampling. TODO: better expand here.

Table 3: Parameter configurations for each dataset: number N_W of wedges to sample for Wedge-Sampling; probabilities p_h, p_t for sampling head and tail nodes, memory budgets B_{M_h}, B_{M_t} for head and tail main samples, B_{A_h}, B_{A_t} for head and tail auxiliary samples, for BOLIDE.

Dataset	N_W	p_h	p_t	B_{M_h}	B_{M_t}	$B_{A_h} = B_{A_t}$
SK	$4k$	0.2	$0.05 \cdot n/m$	$0.09 \cdot m$	$0.01 \cdot m$	$0.05 \cdot m$
LJ	$16k$	0.1	$0.1 \cdot n/m$	$0.04 \cdot m$	$0.01 \cdot m$	$0.01 \cdot m$
ORK	$8k$	0.4	$0.05 \cdot n/m$	$0.04 \cdot m$	$0.01 \cdot m$	$0.05 \cdot m$
TW	$32k$	0.1	$0.1 \cdot n/m$	$0.02 \cdot m$	$0.01 \cdot m$	$0.01 \cdot m$
FR	$32k$	0.05	$0.05 \cdot n/m$	$0.02 \cdot m$	$0.01 \cdot m$	$0.01 \cdot m$

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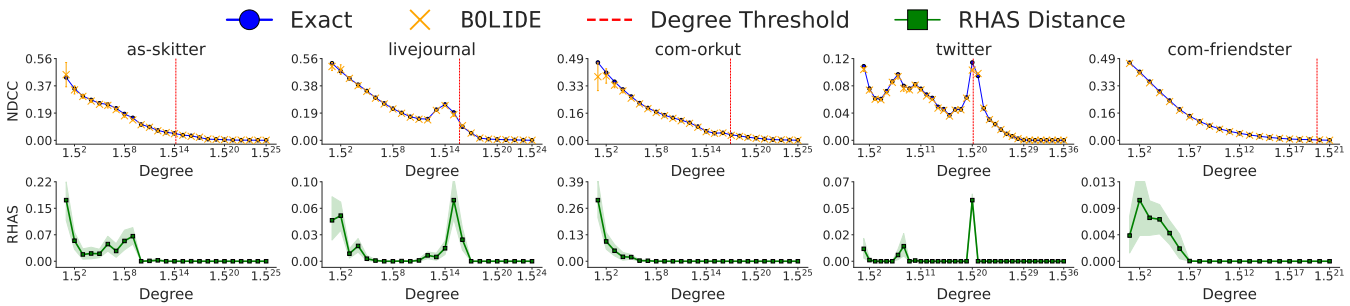

Figure 4: Exact and estimated distributions for NDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.5 (i.e., $D_i = [1.5^i, 1.5^{i+1})$). For each dataset, average and standard deviation over 10 independent repetitions are shown.

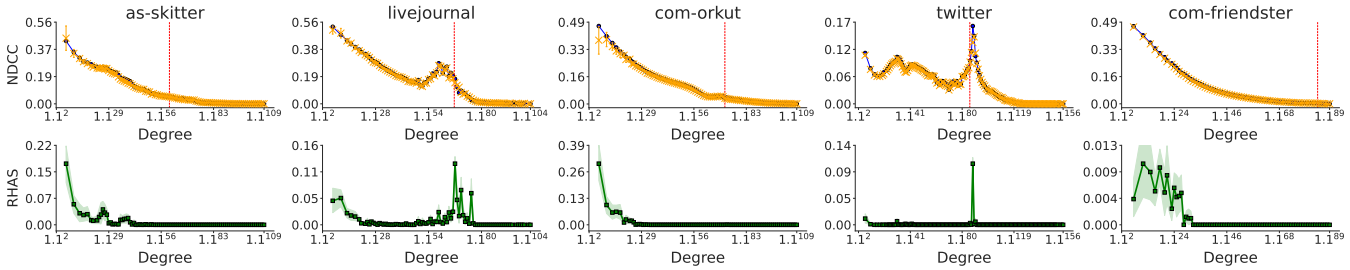

Figure 5: Exact and estimated distributions for NDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.1 (i.e., $D_i = [1.1^i, 1.1^{i+1})$). For each dataset, average and standard deviation over 10 independent repetitions are shown.

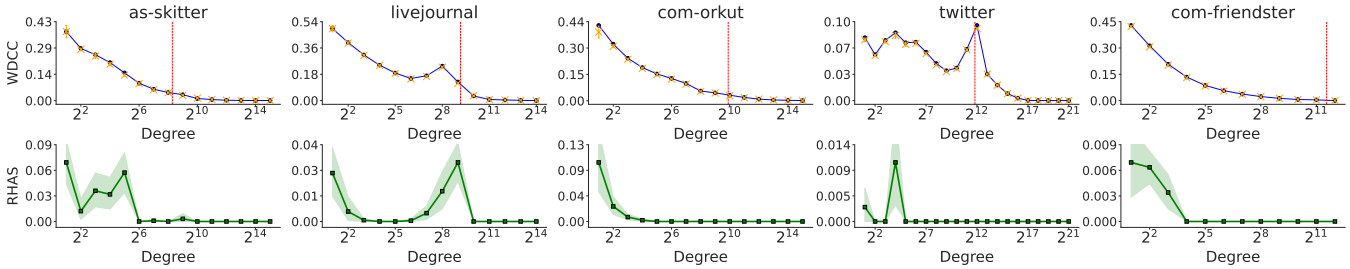

Figure 6: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 2 (i.e., $D_i = [2^i, 2^{i+1})$). For each dataset, average and standard deviation over 10 independent repetitions are shown.

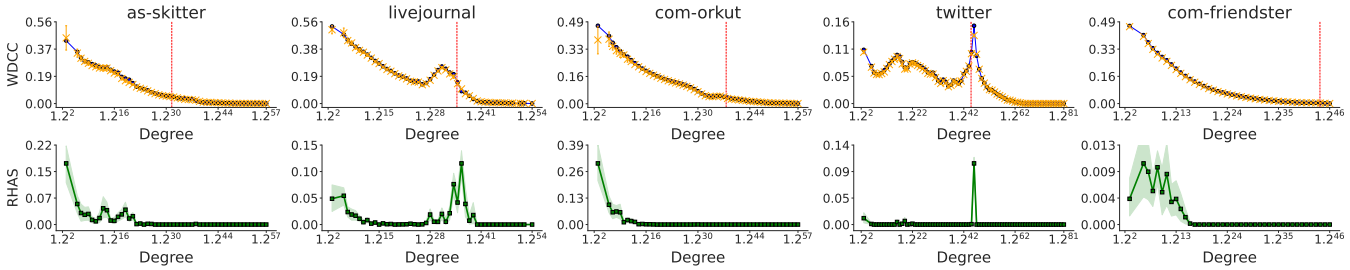

Figure 7: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.2 (i.e., $D_i = [1.2^i, 1.2^{i+1})$). For each dataset, average and standard deviation over 10 independent repetitions are shown.

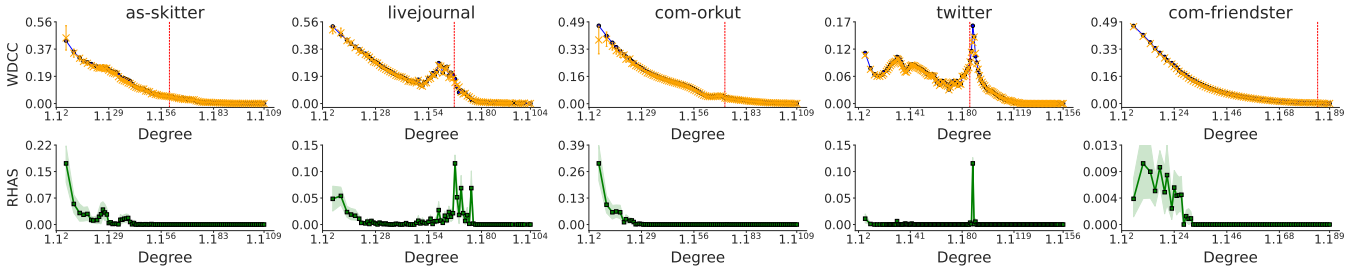

Figure 8: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.1 (i.e., $D_i = [1.1^i, 1.1^{i+1})$). For each dataset, average and standard deviation over 10 independent repetitions are shown.